Omar Mohamed Eid - Week 3

BigBird: A Transformer for Long Documents

Research report, Task 0

1. Introduction

In this report I look at BigBird, a transformer model from Google Research. It was introduced in the paper "Big Bird: Transformers for Longer Sequences" (Zaheer et al., NeurIPS 2020). The main idea is simple: make the attention part of a transformer cheap enough to read very long inputs, like full articles, long contracts or DNA sequences, without losing too much quality.

2. Motivation

Models like BERT are very good at language tasks, but they can only read about 512 tokens at a time. The reason is self-attention. In standard self-attention every token looks at every other token. If the input has n tokens, the model has to build an n x n attention matrix. So the time and memory grow with n squared.

A quick example: going from 512 to 4096 tokens makes the input 8 times longer, but the attention matrix becomes 64 times bigger (8 x 8). On normal GPUs this runs out of memory very fast.

Because of this limit, people usually cut long documents into pieces of 512 tokens and process each piece alone. This is a problem because the model loses the connection between the pieces. For tasks like question answering over a long text or summarizing a long paper, the answer can depend on parts that are far away from each other.

So the goal of BigBird is to keep the strengths of full attention, but make the cost grow linearly with the input length.

3. Architecture

BigBird has the same overall structure as BERT. It is a stack of transformer encoder layers with embeddings, multi-head attention and feed-forward layers. The only big change is inside the attention. Instead of letting every token attend to every token, each token only attends to a small, carefully chosen set of tokens. This is called sparse attention.

If we think of the tokens as nodes in a graph, full attention is a complete graph (everything connected to everything). BigBird uses a sparse graph that still lets information travel quickly from any token to any other token across a few layers.

4. Sparse attention mechanism

BigBird combines three kinds of attention. Each one has a different job.

4.1 Sliding window attention

Each token attends to a few neighbors on its left and right. Language is mostly local, so nearby words are usually the most important ones. This is the same idea used in Longformer. It captures local context.

4.2 Global tokens

A few special tokens attend to all tokens in the sequence, and all tokens attend to them. They work like hubs that collect information from the whole text and spread it back. The paper describes two ways to build them:

- ITC (internal transformer construction): some of the existing tokens (for example the [CLS] token) are made global.
- ETC (extended transformer construction): extra new tokens are added to the sequence and used as global tokens.

Global tokens make sure any two tokens are at most two steps away from each other.

4.3 Random attention

Each token also attends to a few randomly chosen tokens. This is the new part compared to Longformer. A random graph has a short path between any two nodes, so random links work as shortcuts across the sequence. This helps information move quickly over long distances, which the window alone would do slowly.

4.4 Block-sparse implementation

The three patterns above, taken token by token, would be slow on GPUs and TPUs because these chips like dense matrix multiplication and do not like scattered memory access. So BigBird works with blocks. The sequence is split into blocks of tokens (the Hugging Face version uses a block size of 64 by default), and the sparse pattern is defined between blocks, not between single tokens. Each block attends to:

- its neighboring blocks (the window, 3 blocks by default),
- a few random blocks (3 by default),
- the global blocks.

Then the model can compute attention on whole blocks with normal dense matrix operations. This is what makes the idea work in practice.

5. Complexity comparison

Here n is the sequence length, w is the window size, r is the number of random connections and g is the number of global tokens.

Model	Attention type	Time and memory	Typical max length
BERT	Full self-attention	$O(n^2)$	512
Longformer	Sliding window + global	O(n x (w + g)), approx. O(n)	4096
BigBird	Window + global + random	O(n x (w + r + g)), approx. O(n)	4096

The values w, r and g are small constants that do not depend on n, so both Longformer and BigBird are linear in the length. The practical difference between them is not the complexity class. It is the attention pattern. BigBird adds random links, and it comes with a theory behind it. The authors prove that BigBird is a universal approximator of sequence functions and is Turing complete, like a full transformer. They also show that sparse attention can lose some power compared to full attention on certain tasks, which is a trade-off to keep in mind.

6. Suitable applications

BigBird is a good choice when the input is long and the whole text matters:

- Question answering on long documents (the paper reports results on datasets such as HotpotQA, Natural Questions, TriviaQA and WikiHop).
- Summarization of long texts such as scientific papers (BigBird-Pegasus on arXiv and PubMed) and patents.
- Classification of long documents, for example legal or medical documents.
- Genomics. The paper applies it to DNA sequences, for example promoter region prediction, because DNA sequences are very long.
- Search and information retrieval over full documents instead of short passages.

7. Advantages

- Linear cost in sequence length, so it can handle about 8 times more tokens than BERT with similar hardware.
- Keeps a global view of the text through the global tokens, and long-range links through the random attention.
- Has theoretical support (universal approximation and Turing completeness).
- Works as a drop-in replacement for BERT-style encoders, and pretrained models are available in Hugging Face Transformers.

- Works beyond text, for example on genomic data.

8. Limitations

- The sparse pattern is more complicated to implement than full attention, and random access is not very friendly to hardware. The block structure fixes much of this, but it is still more complex than a normal transformer.
- For short inputs there is no benefit. In Hugging Face, BigBird falls back to full attention when the sequence is short, because the block-sparse method only pays off for longer inputs.
- Because tokens do not see everything directly, some information must pass through several layers, so it can be less precise than full attention on some tasks.
- It adds extra settings to tune (block size, number of random blocks, window size, number of global tokens).
- The random pattern is not based on the content of the text, so it may connect tokens that do not matter.
- It still has a length limit (around 4096 tokens in the common pretrained models), so very long documents need to be chunked anyway.

9. Performance

In the paper, BigBird gave better results than RoBERTa-style models that use 512 tokens on tasks with long inputs, mainly question answering and summarization, and it set strong results on several of those benchmarks at that time. It also gave a clear improvement on the genomic tasks. In practice, BigBird and Longformer perform at a similar level on most long-document benchmarks, and the better choice often depends on the task and on which pretrained checkpoint is available. For short text, BERT or RoBERTa are simpler and just as good.

10. Conclusion

BigBird solves the quadratic cost problem of self-attention by replacing it with a sparse pattern made of window, global and random attention, computed in blocks so that it runs efficiently on hardware. The result is a model that scales linearly with the input length and can read much longer documents than BERT while keeping good accuracy. Its main drawbacks are a more complex implementation and no real benefit on short inputs.

References

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